

OMANSHU BHATT

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Portfolio: <https://omanshu-bhatt-portfolio.vercel.app/> Github: <https://github.com/Omanshu09> LinkedIn: <https://www.linkedin.com/in/omanshubhatt/>

PROFESSIONAL SUMMARY

Computer Engineering graduate with experience building and analyzing API-driven products and data-intensive systems. Strong in Python, JavaScript, REST APIs, databases, data validation, debugging and identifying inconsistencies in real-world data. Experienced in turning technical findings into structured, actionable insights through projects and engineering internships.

TECHNICAL SKILLS

Languages: Python, C, TypeScript, JavaScript, Java

Backend & Systems: FastAPI, Node.js, Express, REST APIs, Distributed Systems, Concurrent Systems, System Design

Databases & Storage: PostgreSQL, SQLite, Redis, Elasticsearch, Neon, Data Modeling

Distributed & Realtime: Yjs, CRDTs, Ably, WebSockets, IndexedDB, Distributed State Synchronization

Infrastructure & DevOps: Docker, Kubernetes, Linux, Git, Kafka, CI/CD (GitHub Actions, Jenkins)

AI / ML: PyTorch, Generative AI, RAG, LangChain, Vector Databases

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Object-Oriented Programming

FEATURED PROJECTS

Web Research & Evidence Engine

- Built a Python based web research engine integrating web crawling, content extraction, multi-source retrieval, relevance filtering and ranking to transform unstructured web content into structured, source backed results. Implemented information retrieval and evidence extraction pipelines using BM25, embeddings and Elasticsearch.
 - Tech:** Python, FastAPI, Elasticsearch, Information Retrieval, Web Crawling, Data Processing, RAG, React, Docker
 - Benchmarked across 100+ research queries, achieving ~80% relevant source retrieval with sub-10s median response time.
 - Independently architected and competitively submitted at the Agents for Humans Hackathon | Amazon/AWS
- Code:** <https://github.com/Omanshu09/Search-Engine> **Live:** <https://search-engine-frontend-xi.vercel.app/>

Real-Time Collaborative Engineering Workspace

- Built a real-time collaborative workspace with multi-user CRDT-based Notepad editing, near-instant synchronization, independent Python execution, shared output, and automatic state recovery during connectivity loss.
 - Tech:** React, TypeScript, Vite, Node.js, FastAPI, Express, Yjs, CRDT, Ably, IndexedDB, PostgreSQL, Neon, E2B, REST APIs, Vercel, Render
 - Instant collaborative synchronization during multi user editing, with offline local state retention and isolated Python execution
 - Independently architected and competitively submitted at GatewayHacks 2026 | Devpost
- Code:** <https://github.com/Omanshu09/Nexus-deployment> **Live:** <https://nexus-deployment-frontend.vercel.app/>

Real-Time Translated Video Calling Platform

- Built a video calling platform where each participant speaks their own language while seeing and hearing the other translated into theirs, running Whisper speech recognition and NLLB-200 translation entirely on-device via WebGPU with a WASM fallback.
 - Tech:** React, TypeScript, Vite, WebRTC, Web Workers, Whisper STT, NLLB-200, WebGPU, Web Speech API, Vercel
 - Validated end-to-end across the full curated language set — every language transcribes, translates and dubs correctly with low-latency captioning
- Code:** <https://github.com/Omanshu09/Fluo-Call> **Live:** <https://fluo-call.vercel.app/>

EXPERIENCE

Trainee Intern | Indian Oil Corporation Limited (IOCL) | July 2025 – August 2025

- Worked with operational data and database systems to analyze trends, identify recurring patterns, and support data-backed improvements in engineering processes.
- Collaborated directly with IOCL engineers and technical teams, gaining hands-on experience in database management, enterprise engineering workflows, and quality-focused problem solving.

Virtual Intern — AI/ML & Robotics | Kodacy | July 2024 – August 2024

- Applied Python to AI/ML and data-analysis tasks, working with datasets to extract patterns, evaluate results, and build practical solutions.
- Developed Python-based robotics simulation prototypes, gaining exposure to algorithmic problem solving, automation and intelligent systems.

CERTIFICATIONS

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| • Oracle — Agentic AI Certified Foundations Associate | • Commonwealth Bank — Software Engineering Job Simulation |
| • TCS iON Career Edge - Young professional | • University of Oxford — AI, Justice and rule of law |
| • NASA — Hyperspectral Data for Land and Coastal Systems | • BCGX — GenAI Job Simulation |
| • Deloitte — Cyber Job Simulation | • Cisco — Data Science |

EDUCATION

- Govind Ballabh Pant University of Agriculture and Technology, Pantnagar | 2022 – 2026 | CGPA: 7.2**
Bachelor of Technology (B.Tech) - Computer Engineering
- Beersheba Senior Secondary School, Almora**
Class XII | 2021 – 2022 | 90.4%
Class X | 2019 – 2020 | 95.3